■ Rainbow in a Glass

A Hands-On Kids Science Experiment

What is a Rainbow?

A rainbow is an arc of colours that appears in the sky after rain. When sunlight passes through tiny water droplets, it bends and splits into its different colours. A rainbow has seven colours — **red, orange, yellow, green, blue, indigo,** and **violet** — though in this experiment we will work with six beautiful colours.

- ■■ Rainbows are made by sunlight and rain working together.
- ■ You can't touch a real rainbow — it moves as you move!
- ■ Rainbows have 7 colours, but we use 6 in this experiment.

■ Materials You Need

Water	**Sugar**	**Food Colouring (6 colours)**
Clear Glass	**Spoon**	**Measuring Cup**

■ Step-by-Step Instructions

1 Make Sugar Water

Mix sugar with water in **six separate small bowls** using *different amounts of sugar*. The first bowl gets the most sugar (about 6 teaspoons) and each successive bowl gets 1 teaspoon less, so the last bowl has only 1 teaspoon. Stir each bowl well until the sugar is fully dissolved.

2 Add Food Colours

Add a different food colour to each bowl: **red** → most sugar (heaviest), **orange**, **yellow**, **green**, **blue**, and **purple** → least sugar (lightest). Stir gently to mix the colour evenly.

3 Pour the First Layer

Hold your clear glass steady. Pour the **heaviest** liquid (red — most sugar) slowly into the bottom of the glass first. Go slowly and carefully!

4 Layer the Remaining Colours

Place a spoon inside the glass, just above the red layer, curved side up. Pour each next colour very gently over the back of the spoon so it flows softly without mixing into the layer below. Repeat for orange → yellow → green → blue → purple (least sugar on top).

5 Admire Your Rainbow!

Remove the spoon carefully and look at your glass from the side. You should see **six beautiful, separate colour layers** — a rainbow in a glass! ■

■ The Science Behind It

Why Do the Layers Stay Separate?

Sugar dissolved in water makes the water **denser** (heavier). Denser liquids sink below less dense ones — just like a rock sinks in water. The more sugar a liquid contains, the heavier it is, so it sits at the bottom. The liquid with the least sugar is the lightest and floats to the top. This difference in **density** keeps each coloured layer in place!

Layer	Colour	Sugar Amount	Density
Bottom (1st)	Red	Most (6 tsp)	Heaviest ▼
2nd	Orange	5 tsp	↓
3rd	Yellow	4 tsp	↓
4th	Green	3 tsp	↓
5th	Blue	2 tsp	↓
Top (6th)	Purple	Least (1 tsp)	Lightest ▲

■■ Safety Tips

- Always ask an **adult to help** with this experiment.
- Do **not drink** the coloured water — it is for science only!

• Clean up spills quickly to avoid staining surfaces.

• Wash your hands thoroughly after the experiment.

✏️■ Quick Quiz

Answer these questions after the experiment:

1. Which colour goes at the **bottom** of the glass, and why?

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2. What ingredient makes one liquid heavier than another?

   ____

3. Why do the layers stay separate instead of mixing?

   ____

4. How many colours are in a real rainbow?

   ____

■ Recap

■	Prepare sugar water with six different sugar amounts.
■	Add one food colour to each bowl.
■■	Pour the heaviest (most sugar) liquid into the glass first.
■	Layer the remaining colours one by one over the back of a spoon.
■	Watch the beautiful rainbow layers form in the glass!

Great work, little scientists! ■

You have just discovered the power of density! Keep exploring and asking questions — that is what science is all about.